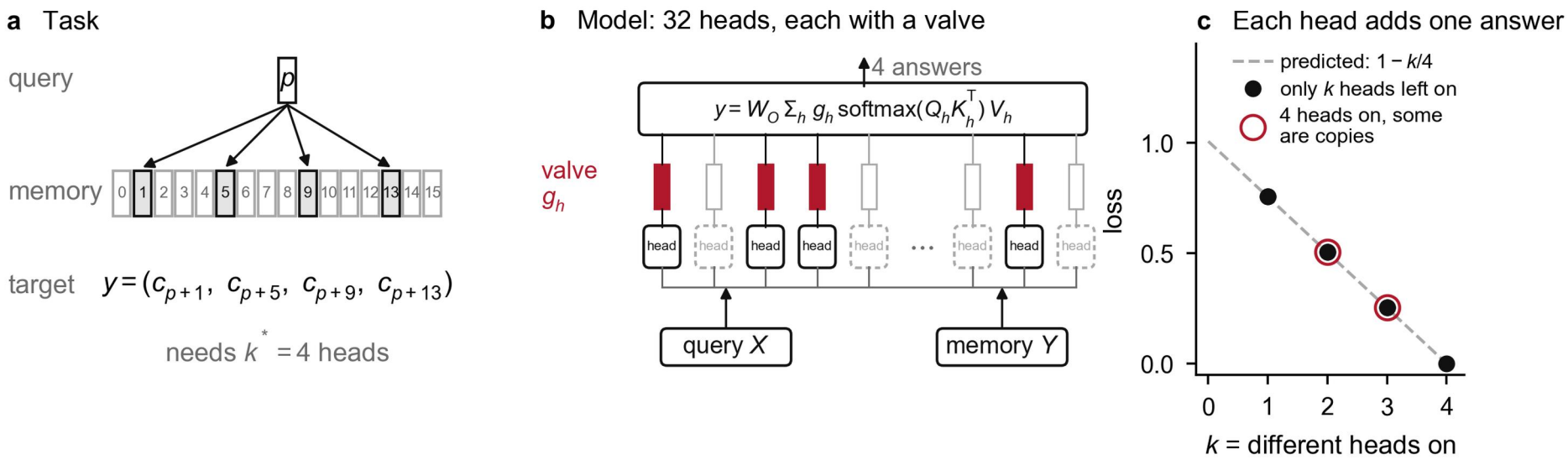
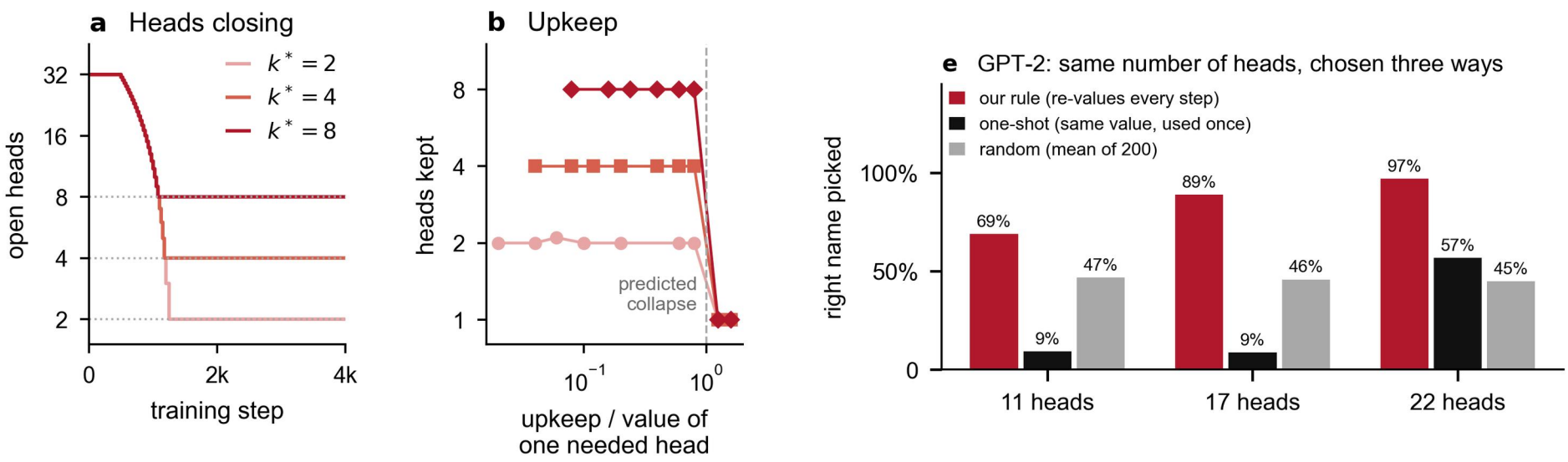
1 What it is: every head has a valve and pays rent; close it when it's worth less than its rent

The rule is the step-by-step minimiser of loss + rent × open heads (Murray's law for vessels).



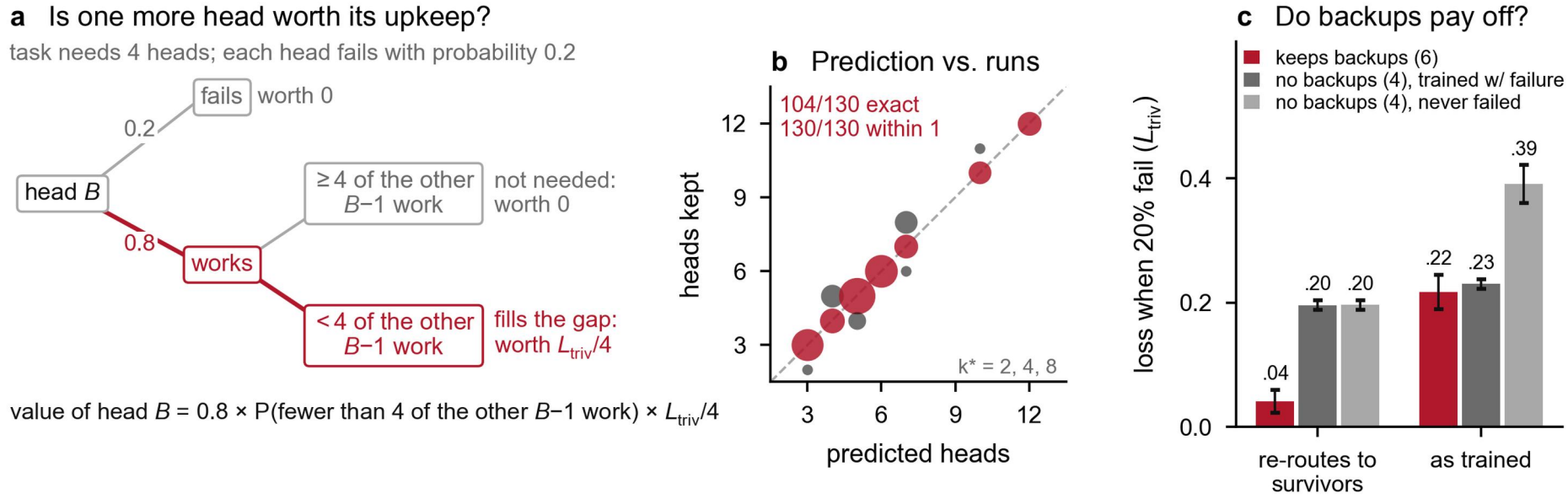
2 Keeps exactly the heads needed, collapses where theory says; beats one-shot on GPT-2

29/30 exact, never broke the model (3 published methods broke it every time); 36/36 at the predicted collapse. e: same value used once picks 9%; ours 89%.



3 It predicts how many spare heads to keep when heads can fail, before training

Predictions written down first: 79/100 exact, 100/100 within one head. Spares cut the loss 5x when heads fail.



4 It recovers circuits found by hand

Induction 10/10 runs. GPT-2: core of 11 heads all in the published circuit; one head kept at every stage; nested.

